Encoding hierarchical conformational and compositional heterogeneity in mmCIF/PDBx

Executive Summary

PDBx/mmCIF models do not adequately encode conformational and compositional heterogeneity intrinsic in structural biology data. This problem becomes more pressing when examining new classes of experiments that frequently or expressly induce such heterogeneities, including fragment screening and time-resolved techniques. Leveraging mmCIF's expressive nature, we propose a new category that enables users and software to explicitly describe heterogeneity by explicitly encoding the relationships between different pieces of heterogeneity.

The Problem

Most PDB depositions result from X-ray crystallography or single-particle cryo-electron microscopy (cryo-EM), which collect information from thousands to trillions of copies of the macromolecular complex. Each copy of the macromolecular complex can vary in the relative positions of atoms within it (conformational heterogeneity) or in the presence/absence of part of the complex (compositional heterogeneity). Conformational heterogeneity can occur across length scales from a single atom (e.g. a terminal hydroxyl in serine partially occupying two locations) to entire domain movements (e.g. the ribosome stalk partially occupying two locations)[1–6]. Compositional heterogeneity can also occur across length scales from an ion to a protein subunit[7–11]. Models intended to describe this experimental data must be able to effectively capture this inherent conformational and compositional heterogeneity; however, the current encoding falls short.

Currently, in PDBx/mmCIFs, harmonic conformational heterogeneity is captured by B-factors or translation-libration-screw [TLS] parameters. In contrast, non-harmonic conformational or compositional heterogeneity is captured through the alternative location indicator (altloc). Altlocs indicate the mutually exclusive positions any given atom can occupy, providing information on coordinates, occupancy, and B-factors for each position. Multiple atoms are often grouped into the same altloc, indicating that, on a local level, they are part of the same state.

This convention falls well short when it becomes necessary to describe how altlocs are related to each other[12]. It has long been recognized that additional information on the relationships *between* altlocs is necessary to adequately fit this data[13,14]. However, this information is only provided in refinement programs (via grouped occupancy) and is not explicitly stated in deposited models.

For example, conditional on the presence or absence of a ligand, two ethylene glycol (EDO) molecules may be present, one of which can occupy two mutually exclusive states (**Figure 1a**). While there are ways to apply restraints and constraints in refinement, this conditionality cannot be expressed in PDBx/mmCIF altlocs. This limitation hurts our ability to read a model and understand the downstream

Biological Applications

While the interplay between conformational and compositional heterogeneity is present in all structural experiments, accurate encoding has particularly large implications for several classes of

experiments that frequently or explicitly induce such heterogeneities, including fragment screening and time-resolved techniques.

Prominent use cases include:

X-ray crystallography fragment screening: X-ray crystallography fragment screening is an increasingly popular drug discovery technique that involves collecting hundreds to thousands of crystal structures of small-molecule fragments bound to proteins. Collectively, these structures provide building blocks for ligands by observing and modeling diverse small molecule starting points[15]. The challenge arises because the fragment or ligand is bound to only a subset of macromolecules, inducing partial occupancy changes across states, resulting in experimental data with high compositional and conformational heterogeneity [10,16,17]. Such screens make up an increasing fraction of depositions in the PDB and are of significant importance to the pharmaceutical industry. A clear understanding of the hierarchical heterogeneity is essential for future chemical modifications and for using these data in automated methods or machine learning.

Time-resolved experiments: Time-resolved X-ray crystallography or cryo-EM involves collecting multiple datasets over time as the sample undergoes a dynamic process. To achieve this, the sample (typically protein crystals) is rapidly subjected to a specific stimulus (e.g. with temperature, light, or electric field). The changes induce structural rearrangements in the protein, and diffraction data are collected at multiple time points, enabling real-time tracking of these changes. These studies are critical to understanding protein function in terms of time-domain structural heterogeneity, such as catalysis, ligand binding, or macromolecular machines [18–22]. While experiments often yield multiple depositions linked to different maps, within a single deposition, the ground state contains a mixture of conformers, and the excited state also contains a mixture of conformers, although we expect different occupancies. The inability to express the dependencies among occupancy groups in the mmCIF/PDBx means that the structure can be understood only with the accompanying literature, which is problematic for automated methods and machine learning.

Other experiments purposefully seek to enrich compositional heterogeneity. In these cases, a single collection reflects data from many biochemically distinct samples. For example, a collection of small molecules in a “cocktail” can be soaked into the same protein crystal, resulting in a single protein bound to many ligands - potentially in the same binding site [23]. In another example, native complexes with multiple binding partners can be purified for Cryo-EM [24] or imaged by cryo-electron tomography (cryo-ET) [6,25].

Proposed encoding of heterogeneity in PDBx/mmCIF

Thanks to the more expressive data model of PDBx/mmCIF, it is now possible to explicitly encode heterogeneity-related relationships that are implicit or ambiguous in current representations. To enable this, we propose a new category that directly encodes (i) the hierarchical relationships among compositional and conformational states and (ii) the coexistence or mutual exclusivity of those states.

Together, these two tables provide a formal mechanism for representing both the dependency structure and logical compatibility among heterogeneous states in multiconformer or ensemble models.

The core table, termed the heterogeneity hierarchy, is a mandatory loop that links heterogeneity identifiers associated with atoms in the `_atom_site` category (and, when applicable, model IDs) to one another through explicit parent–child relationships. This hierarchy encodes how individual heterogeneous states are related—such as alternative ligand-binding events, conformational substates, or compositional variants—allowing users and software to reconstruct a structured view of heterogeneity rather than relying on proximity-based altloc conventions alone. The hierarchy table is complemented by a second table, the state coexistence table, which explicitly specifies which heterogeneity states may or may not co-occur.

Within the `_atom_site` category, we introduce a heterogeneity ID. This identifier is optional for PDB entries that do not require explicit heterogeneity encoding, but becomes mandatory for atoms that participate in the heterogeneity hierarchy. In some cases, heterogeneity IDs may correspond directly to existing altloc identifiers; however, unlike altlocs, each heterogeneity ID must represent a distinct biochemical or structural state. For example, two spatially distant atoms may both be labeled with altloc “A” under current conventions, even though there is no evidence that they represent the same underlying state. In such cases, distinct heterogeneity IDs would be assigned, allowing their relationships (or lack thereof) to be represented unambiguously within the hierarchy.

The heterogeneity hierarchy category uses four fields: `name`, `id`, `parent`, and `details`. The `name` field is mandatory and provides a short, human-readable label for the state. The `id` field is the category key and serves as the formal link between this table and the `_atom_site` category. The optional `details` field allows descriptive text (quoted if it contains spaces). The `parent` field references another heterogeneity ID and encodes the hierarchical structure: each heterogeneity ID has at most one parent, whereas any parent may have multiple children. This constraint enforces a tree-like hierarchy that captures how states refine or branch from one another. By traversing parent–child relationships, software can explicitly reconstruct the hierarchy of heterogeneity encoded across the atomic model.

Unlike traditional altloc usage—where identical altloc identifiers implicitly define a single mutually exclusive state within a local clash radius—the heterogeneity hierarchy provides a global, explicit description of state relationships. When conformational or compositional states are known to be linked biochemically, physically, or mechanistically, they may share the same altloc and heterogeneity ID. Conversely, when no such linkage is known, distinct heterogeneity IDs are assigned, even if the altloc labels coincide. This ensures that altloc usage remains compatible with existing practice while heterogeneity IDs provide the formal semantic layer needed to interpret relationships correctly.

Mutual compatibility among states is encoded separately in the state coexistence table. This table operates on heterogeneity IDs defined in the hierarchy and specifies logical rules(e.g., OR or NOT) that determine whether states may co-occur. Hierarchical relationships inherently encode `AND`

relationships (parent and child states are coexistent by definition), whereas the coexistence table captures non-hierarchical logical constraints. Importantly, if no coexistence rule is specified between two heterogeneity IDs, they are assumed not to be mutually exclusive. As with altlocs, individual heterogeneity IDs are not expected to represent complete models in isolation; rather, the full hierarchy and coexistence definitions together should account for all atoms and fully explain the experimental data.

EXAMPLE LOOPS:

```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name      .    Base   ?
EDO1_bound     Base B   ?
Fragment_bound Base C   ?
EDO2_bound     Base D   ?
EDO3_bound     Base E   ?



loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment
   id rule heterogeneity_id heterogeneity_ids description
    1  NOT  B               C    EDO1_bound_NOT_Fragment
    2  OR   C               D,E  Fragment_OR_(EDO2, EDO3)
    3  OR   B               D,E  EDO1_OR_(EDO2, EDO3)
    4  NOT  D               E    EDO2_bound_NOT_EDO3_bound

Example (cofactor + substrate in the second binding site): (EDO1 OR Fragment) AND (EDO2 OR EDO3)
```

Examples

Consider a common situation in fragment screen data with extensive compositional heterogeneity. This structure has four ligands: three EDO molecules and a small fragment bound (**Figure 1A**). Based on occupancy, biochemical information, and overlap, it is known that when the fragment or EDO1 is bound, either EDO2 or EDO3 can then be bound. When EDO2 or EDO3 is bound, either EDO1 or the fragment is bound (**Figure 1B**). This example is shown in the proposed heterogeneity table in Figure 1C, and Figure 1D displays the atom table (which shows only one atom per ligand for simplicity).

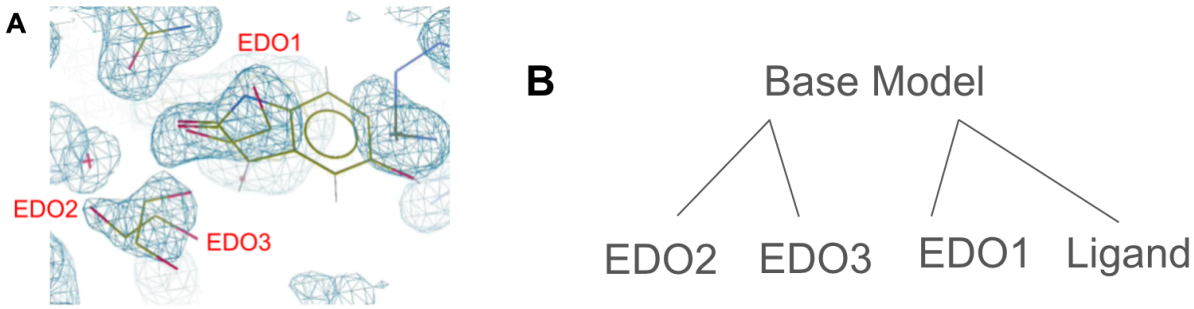


Figure 1. A representation of simple compositional heterogeneity is proposed. (**A**) Overlay of the compositional heterogeneity in the experimental data. (**B**) Tree representation of how compositional heterogeneity is related to each other.

```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name       .    Base   ?
EDO1_bound     Base B      ?
Fragment_bound Base C      ?
EDO2_bound     Base D      ?
EDO3_bound      Base E      ?


Loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment
   id rule heterogeneity_id heterogeneity_ids description
    1  NOT  B               C    EDO1_bound_NOT_Fragment
    2  OR   C               D,E  Fragment_OR_(EDO2, EDO3)
    3  OR   B               D,E  EDO1_OR_(EDO2, EDO3)
    4  NOT  D               E    EDO2_bound_NOT_EDO3_bound

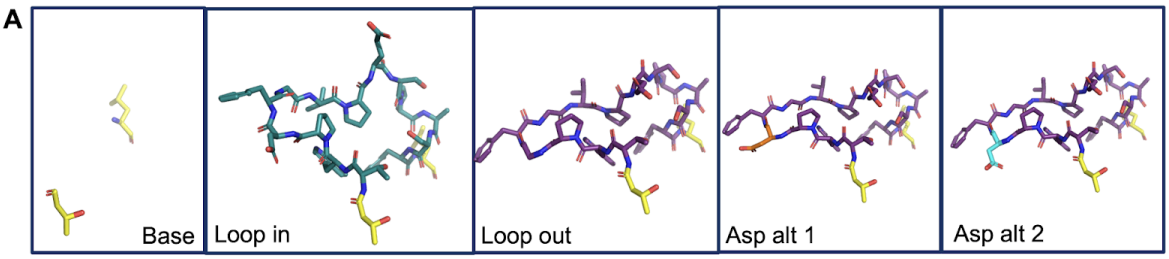

loop_
_atom_site.group_PDB
_atom_site.id
_atom_site.type_symbol
_atom_site.label_atom_id
_atom_site.label_alt_id
_atom_site.label_comp_id
_atom_site.label_asym_id
_atom_site.label_entity_id
_atom_site.label_seq_id
_atom_site.pdbx_PDB_ins_code
_atom_site.Cartn_x
_atom_site.Cartn_y
_atom_site.Cartn_z
_atom_site.occupancy
```

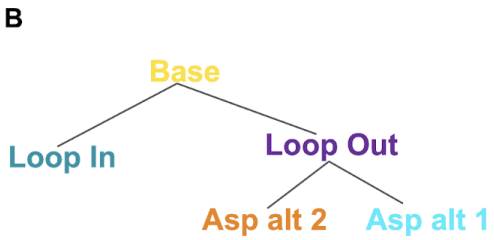
```
_atom_site.B_iso_or_equiv
_atom_site.pdbx_formal_charge
_atom_site.auth_seq_id
_atom_site.auth_comp_id
_atom_site.auth_asym_id
_atom_site.auth_atom_id
_atom_site.pdbx_PDB_model_num
_atom_site.pdbx_heterogeneity_id

  ATOM 1 C C B LIG   A 1 ? 2.58600 -21.06800 20.07200 0.500 14.77 N 1 C
  ATOM 2 C C A EDO1  A 2 ? 3.66000 -20.86300 21.26100 0.500 17.96 C 1 B
  ATOM 3 C C C EDO2  A 3 ? 4.98500 -21.56000 20.99100 0.500 17.25 C 1 D
  ATOM 4 C C D EDO3  A 4 ? 5.03800 -22.59100 20.31200 0.500 18.69 O 1 E
```

In the heterogeneity loop (**Figure 1C**), structures will default to start with a base hierarchy ID, including all atoms with only a single set of coordinates (i.e. no altlocs) or those in the model one. The user can amend this as needed. The two ‘children’, EDO1 and fragment states, come from the base, each associated with its altloc ID (A and B, respectively). The EDO1 bound state then has two children, EDO2 and EDO3. Each of these would have its own altloc ID (C, D), and both would have the parent of EDO1. The hierarchy loop aims to be read from the bottom up, with children encompassing their parents. For example, reading the EDO3 state – this state would include all positions of atoms in the EDO3 state, but also the EDO1 and base state.

This method also supports the description of hierarchical structures with only conformational heterogeneity (Figure 2; Supplementary File 1) or with both conformational and compositional heterogeneity (**Figure 3**). Figure 2’s example shows only conformational heterogeneity, consisting of a protein loop region in which the backbone splits, representing loop-out and loop-in conformations. Additionally, on the loop-out conformation, an aspartic acid has multiple conformations (**Figure 2B/D**). This is represented in the heterogeneity loop by having the alt_asp as a child of the loop-out conformation (**Figure 2E**). In an example of compositional and conformational heterogeneity, we have three states: a base state, a bound state, and an unbound state (**Figure 3A**). This can be represented in three lines in the loop (**Figure 3B**).





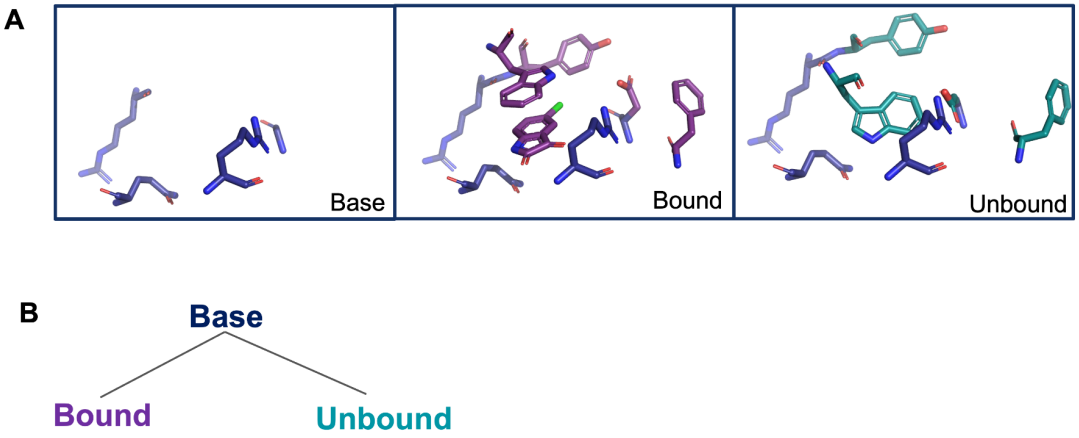

```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name     .         .  ?
Loop_out      Base      A  ?
Loop_in       Base      B  ?
Asp_Alt1      Loop_out  C  ?
Asp_Alt2      Loop_out  D  ?


loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment

id rule heterogeneity_id heterogneiety_ids description
1 NOT Loop_in Loop_out Loop_in_NOT_Loop_out
2 OR  Loop_in Loop_out Loop_in_OR_Loop_out
3 OR  Loop_in Asp_alt_1,Asp_alt_2 Loop_in_or_Asp_alt1_or_Asp_alt2
```

Figure 2. A representation of conformational heterogeneity. (**A**) A stick representation of each state as represented in the heterogeneity loop and tree representation. (**B**) Tree representation of how conformational heterogeneity is related to each other.




```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name    .       Base ?
Bound        Base    A    ?
Ubound       Base    B    ?

loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment

id rule heterogeneity_id heterogneiety_ids description
1. NOT Bound Unbound Bound_NOT_Unbound
```

Figure 3. (**A**) A representation of conformational and compositional heterogeneity is represented in the proposed representation. (**B**) Tree representation of how compositional and conformational heterogeneity are related to each other.

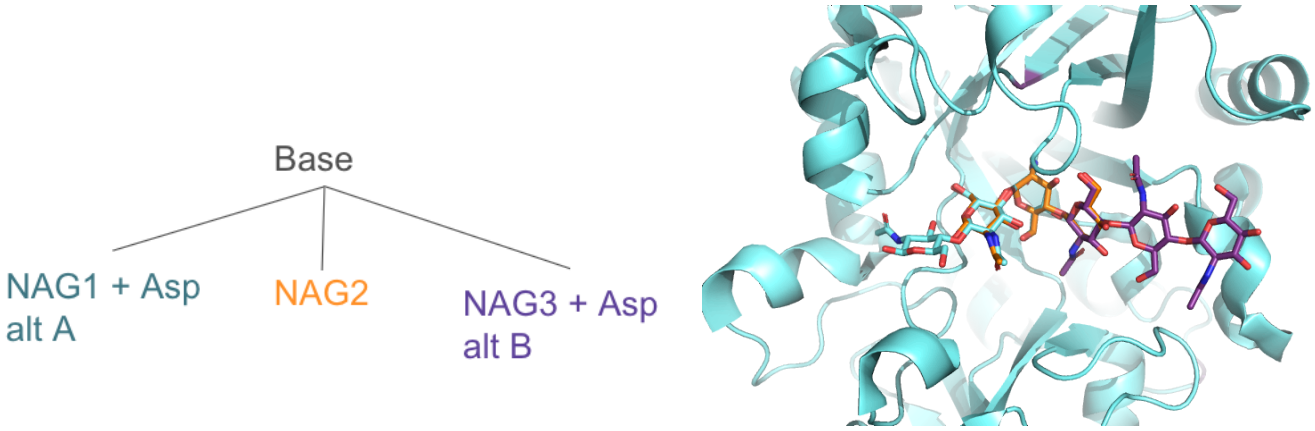


Figure 4. (**A**) Tree representation of how compositional and conformational heterogeneity are related to each other. (**B**) A representation of conformational and compositional heterogeneity is represented in the proposed representation.

```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name       .    Base   ?
NAG1_Asp_A     Base A      ?
NAG2           Base B      ?
NAG3_Asp_B     Base C      ?


loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment
   id rule heterogeneity_id heterogeneity_ids description
       1 NOT  A B  NAG1_not_NAG2
       2 NOT  B C  NAG2_not_NAG3
```

PDB ID 7HHS illustrates a fragment-screening experiment in which both conformational and compositional heterogeneity are present and tightly coupled. The model comprises an apo-protein conformation (shown in green) and a distinct bound-protein conformation (blue). Only the bound conformation can accommodate a ligand, which appears in two mutually exclusive alternative states: ligand alt A (blue) and ligand alt B (pink). As a result, the apo and bound protein states are mutually exclusive and cannot co-exist within the same molecular instance. Likewise, the apo protein cannot co-exist with either ligand alt A or alt B. In contrast, the bound protein can co-exist with either ligand alt A or ligand alt B, but these two ligand states are themselves mutually exclusive and never co-occur. Together, this model captures a hierarchy of heterogeneity in which ligand binding is

conditional on the bound protein conformation, and alternative ligand poses represent exclusive substates within that bound ensemble.

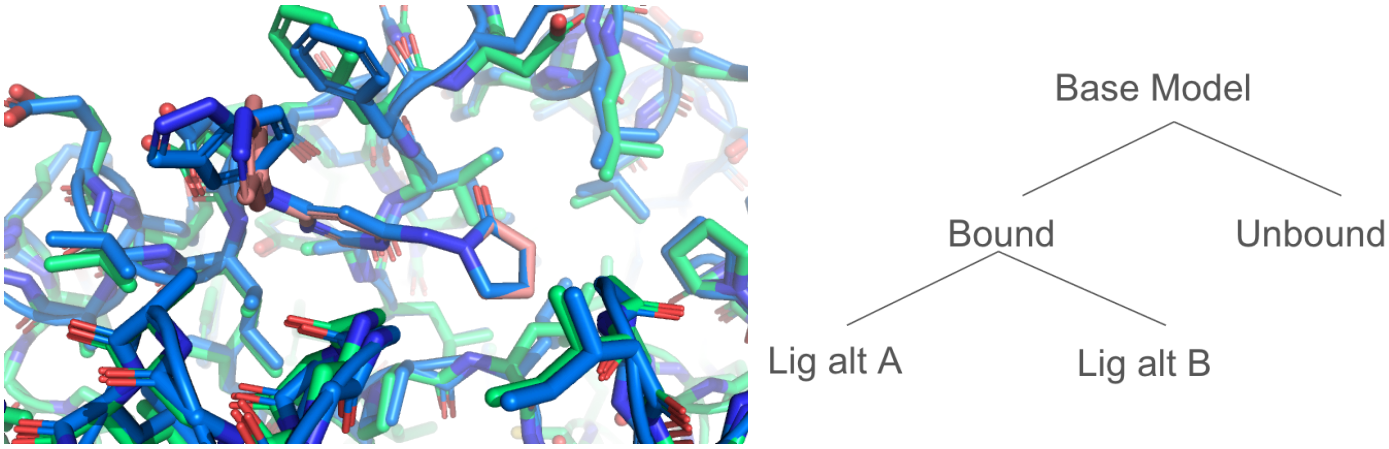


Figure 5. (**A**) A representation of conformational and compositional heterogeneity is represented in the proposed representation. (**B**) Tree representation of how compositional and conformational heterogeneity are related to each other.

```
loop_
_pdbx_heterogeneity_hierarchy.name ---> mandatory
_pdbx_heterogeneity_hierarchy.parent ---> pointer to another _pdbx_heterogeneity_hierarchy.name
_pdbx_heterogeneity_hierarchy.id ---> category key, mandatory
_pdbx_heterogeneity_hierarchy.details

Base_name            .     Base   ?
Unboound             Base  A    ?
Bound                Base  B    ?
Ligand_alt_A         Bound C    ?
Ligand_alt_B         Bound D    ?

loop_
_pdbx_state_coexistence.id
_pdbx_state_coexistence.rule ---> AND/OR/NOT; Default is OR; AND is encoded in the hierarchy;
_pdbx_state_coexistence.heterogeneity_id ---> pointer to _pdbx_heterogeneity_hierarchy.id
_pdbx_state_coexistence.heterogeneity_ids ---> comma separated _pdbx_heterogeneity_hierarchy.ids
_pdbx_state_coexistence.description ---> generic text comment
   id rule heterogeneity_id heterogeneity_ids description
       1 NOT  A B,C,D   Unbound_not_bound_nor_ligaltA_nor_ligaltB
       2 NOT   C D      ligaltA_or_ligaltB
```

Situations not addressed by the heterogeneity category

We deliberately chose not to encode the following scenarios in our current model.

- We excluded a many-to-many relationship, specifying that a parent could have multiple children, but a child could not be associated with multiple parents. This does not strike us as a scientifically sensible thing to do: it would hardcode the commonality between conformations,

whereas rationalising such correlations is decidedly a downstream task, namely the analysis of what the models mean, and highly dependent on the questions asked.
- We did not include the association of multiple maps, such as those used in time-resolved or classification methods in cryoEM. It is conceivable that a single model, supported by multiple maps, could be created within this hierarchical framework in the future.
- The current encoding of hierarchical compositional and conformational heterogeneity is based solely on thermodynamic ensembles without accounting for timescales. However, the framework is flexible enough that timescale information could be incorporated into the table, such as with time-resolved techniques.

Working with existing software

As the proposed loop does not change the atom_site category, it should not break any existing software. However, to fully realize the potential of this encoding, it should be used by existing structural biology software, including model building[10,27–30], refinement programs[31–33], and visualization software[30,34]. Overall, these programs need to account for the heterogeneity loop when present. Additionally, software should account for the possibility that models using the new category may interpret altlocs slightly differently, with each altloc representing a distinct state.

Model Building

For modeling building software, the software should, to the best of its ability, output the loop and/or a grouped occupancy file describing how altlocs are related. While this loop/grouped occupancy may need to be manually adjusted for local heterogeneity hierarchy, it can be automated when clashes determine labeling, as is currently done in qFit and Coot[28,30]. Determining distally hierarchical relationships cannot be done using information from atom_site and will require external methods and individual input from modelers. Additionally, model-building software should err on the side of assigning new altlocs (_atom_site.label_alt_id), unless there is evidence two pieces of heterogeneity are linked. In Coot, we imagine a user-defined flag (i.e. use hierarchy category) for assigning altlocs. Coot would also need to re-assign altlocs as altlocs are deleted from the model during manual building. If the hierarchy category exists in the input model, Coot must amend the table as needed. Currently, this would be based solely on user input and/or conflicting information regarding the conformations.

We also encourage the development of tools that enable modelers to specify the hierarchical relationships between altlocs through a grouped occupancy file or by manually creating the hierarchy loop, with the software automatically correcting the labeling of altlocs in the atom table.

Refinement

We propose that refinement software include a flag to enable the use of information from the hierarchy category. If the flag is enabled, the software should expect a grouped-occupancy file and/or a hierarchy category within the PDBx/mmCIF file. Refinement software should then generate constraints and/or restraints based on the hierarchy or grouped occupancy file or hierarchy category. If a grouped occupancy file is present, it will overwrite the existing hierarchy category in the PDBx/mmCIF file.

Refinement should read the hierarchy loop from children to parents and/or use a grouped occupancy file to determine which hierarchy levels should be equal to or less than one occupancy. Similar to Refmac's current "occupancy group alts complete" parameter, we envision a modified version of this command that would create a restraint ensuring that the occupancy of specific IDs sums to that of a specified parent, reflecting their hierarchical relationship. This would be in addition to the restraint of having atoms with the same ID (pdbx_heterogeneity_hierarchy.id) having the same occupancy. In Phenix and SHELX[31,35], while you can create constraints of mutually exclusive groups using grouped occupancy or PART parameters, it is currently impossible to encode the hierarchy between these groups. When flagged, refinement software must output the category in the PDBx/mmCIF file, specifying how heterogeneity hierarchy is encoded and refined within the model.

Visualization
Visualization software should read the hierarchical heterogeneity loop to display hierarchical heterogeneity. Currently, Pymol supports the use of altlocs in selection criteria and visualization. We aim for Pymol to support the new heterogeneity loop directive in selection and visualization, allowing users to select a state and display all its children and/or parents.

Future Indications

This proposal is primarily motivated by X-ray crystallography, which enables the precise detection of individual states owing to its high resolution and recent technological advances. While cryo-EM currently captures most of its compositional and conformational heterogeneity through 3D classification of maps36, we are beginning to routinely obtain structures at sufficiently high resolution to capture heterogeneity through both map classification and the encoding of multiplicity within structural models.

Furthermore, the machine-readability and human interpretability of the inherent heterogeneity within models are critical for advancing the use and prediction of conformational ensembles. By implementing this model, we anticipate that it will yield new tools and analyses to examine heterogeneity within and across structures, significantly expanding our understanding of the inherent heterogeneity in experimental structural biology data.

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